

Find-S Algorithm

Python Code

```
# Dataset: [Education, Communication, Experience, Selected]
data = [
    ['UG', 'Good', 'Yes', 'Yes'],
    ['UG', 'Good', 'No', 'Yes'],
    ['PG', 'Poor', 'Yes', 'No'],
    ['UG', 'Average', 'Yes', 'Yes']
]

# Initialize Specific Hypothesis
S = ['0'] * (len(data[0]) - 1)

# Find-S Algorithm
for row in data:
    x, y = row[:-1], row[-1]

    if y == 'Yes':
        if S[0] == '0':
            S = x.copy()
        else:
            for i in range(len(S)):
                if S[i] != x[i]:
                    S[i] = '?'

print("Final Specific Hypothesis (S):", S)
```

Code Screenshot

```

1  # Dataset: [Education, Communication, Experience, Selected]
2  data = [
3      ['UG', 'Good', 'Yes', 'Yes'],
4      ['UG', 'Good', 'No', 'Yes'],
5      ['PG', 'Poor', 'Yes', 'No'],
6      ['UG', 'Average', 'Yes', 'Yes']
7  ]
8
9  # Initialize Specific Hypothesis
10 S = ['0'] * (len(data[0]) - 1)
11
12 # Find-S Algorithm
13 for row in data:
14     x, y = row[:-1], row[-1]
15
16     if y == 'Yes':
17         if S[0] == '0':
18             S = x.copy()
19         else:
20             for i in range(len(S)):
21                 if S[i] != x[i]:
22                     S[i] = '?'
23
24 print("Final Specific Hypothesis (S):", S)

```

```

...Final Specific Hypothesis (S): ['UG', '?', '?']

```